1. Measure of Central Tendency

1.1. Mode

- Ungrouped data: mode is the most frequently occurred value
- Grouped data: modal class is the class with highest frequency density
- Not suitable if the data values are very varied

1.2. Median

- Middle value when data is ordered
  - If n odd, median $= \frac{1}{2}(n+1)^{\text{th}}$ value
  - If n even, median $= \frac{1}{2}n^{\text{th}}$ value
- Not affected by extreme values

Estimating Median from Grouped Frequency Table:

$\mathbf{x}$	Frequency $\mathbf{f}$	Cumulative Frequency
$10-20$	4	4
$20-25$	8	12
$25-35$	5	17
$35-50$	3	20

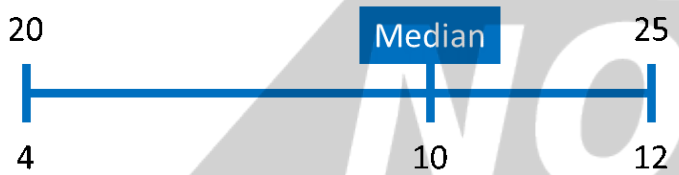
Solution:

Use cumulative frequency to find the middle value i.e.

$$20 \div 2 = 10$$

$\therefore$ *you are finding the 10th value*

The 10th value lies between 20 and 25



$$(12-4)\ :(25-20)$$

$$(12-10)\ :(25-Median)$$

$$25-Median = \frac{12-10}{12-4} \times (25-20)$$

$$Median = 23.75$$

1.3. Summation notation

- The summation notation is used throughout the syllabus to denote the sum of data values or probabilities.

$$\sum x_i = x_1 + x_2 + x_3 + \cdots + x_n$$

- This is the sum of all x values till the nth value.

It should also be known that:

$$\sum a = a + a + a + a + \cdots + a = an \quad [\text{Where } a \text{ is a constant}]$$

- The sum of a constant number n times is $a \times n$.

It is also possible to re-write a summation:

$$\sum (x-a) = \sum x - \sum a = \sum x - an$$

1.4. Mean

- Sum of all data values divided by number of values

$$\overline{x} = \frac{\sum x_i}{n}$$

or

$$\overline{x} = \frac{\sum x_i f_i}{\sum f_i}$$

- Important as it uses all the data values
- Disadvantage: affected by extreme values
- If data is grouped – use the mid-point of group as x

Coded mean

- Coded mean can be thought of as a transformation of all the data values in a data set. It is done to make computation easier.

Coded mean of ungrouped data

$$\overline{x} = \frac{\sum (x-b)}{n} + b$$

Coded mean of grouped data

$$\overline{x} = \frac{\sum (x-b)f}{f} + b$$

- Where f is the corresponding frequency to a value x

Grouped Mean

- The mean of 2 sets of data, or grouped mean, can be found by adding all the data values and the number of values.

Suppose a data set x exists where the sum of the data is $\sum x$ and the number of values is n_x , and another data set y exists where the sum of the data is $\sum y$ and the number of values is n_y .

The mean of this data set is:

$$\text{Grouped mean } = \frac{\sum x + \sum y}{n_x + n_y}$$

Example 1

A summary of 40 values of x gives the follow information:

$$\sum (x-k) = 520$$

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Where k is a constant. Given that the mean of these 40 values of x is 34, find the value of k.

Answer

As we know the value of $\sum(x-k)$:

$$\frac{\sum(x-k)}{40}+k=34 \implies \frac{520}{40}+k=34$$

Now we can solve for k

$$k=34-\frac{520}{40}=21$$

Example 2

For n values of of the variable x, it is given that:

$$\sum(x-50)=144 \text{ and } \sum x=944$$

Find the value of n.

Answer

Using the properties of the summation, we can re-write this as

$$\sum x-50n=144$$

As we know the value of $\sum x$

$$944-50n=144 \implies 50n+144=944$$

So we can find the value of n as

$$50n=800 \implies n=\frac{800}{50}=16$$

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